

Julen Gamboa

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PROFESSIONAL SUMMARY

AI/ML engineer who operationalises model-driven systems from ambiguous requirements through architecture, implementation, testing, release, and adoption. Experienced across agentic applications, model-serving APIs, data engineering, MLOps/LLMOps, RAG/GraphRAG, workflow orchestration, agentic harness engineering, LLM evaluation, LLM fine-tuning, and scientific computing.

WORK EXPERIENCE

Department for Education (DfE), United Kingdom

Jan 2024 to Present

HEO Data Scientist, Government Operational Research Service

Swindon, United Kingdom

- Technical lead across ML/AI, data engineering, MLOps, applications, migration, and governance. Took ambiguous, high-stakes systems through architecture, implementation, testing, release, and adoption while coordinating delivery and coaching colleagues.
- Led development and migration of a benefits-realisation agent that cut a week of manual processing to minutes. Re-engineered it from Copilot Studio into modular Azure DevOps workflows with schema-constrained extraction, bounded generation, Databricks model serving, Microsoft Graph/SharePoint integration, validation, and retries. Managed implementation by delegating to colleagues seeking to learn LLM techniques.
- Authored and published the agentic-development-playbook. Led demonstrations, onboarding, and Copilot CLI adoption across Operational Data Science and the wider DfE analytical community.
- Led the rebuild of a legacy risk-assessment pipeline into governed production. The system serves over 60 colleagues and has supported substantial recovery of public funds; it uses calibrated XGBoost, feature management, MLflow lineage, explainability, drift monitoring, network-graph connectivity-risk methods, CI/CD, and controlled deployment. Led release of its investigator-facing frontend.
- Led migration of a business-critical budget-forecasting system to Python and governed Databricks within an extremely compressed deadline, adding PySpark orchestration, Unity Catalog, MLflow, and MLOps.
- Architected and led a cross-directorate platform supporting over 300 operations colleagues with a comprehensive assessment of provider-closure risk and the need for suitable alternative provision, replacing multiple legacy tools with a Databricks-centred stack including Dash, SQL analytics, OCR/RAG, and reproducible report artifacts.
- Developed FastAPI GraphRAG infrastructure for Copilot CLI and other AI interfaces, measuring between 76% and 99% lower token consumption against baseline usage.

Aramayo Lab, Department of Biology, Texas A&M University

Jan 2023 to Present

AI Consultant, Freelance

College Station, Texas, United States; remote

- Lead two complementary AI workstreams: a Scientific Workbench for research operations and an AI-supported teaching and evaluation platform for BIOL 213 Molecular Cell Biology.
- Built and continue to extend the Scientific Workbench by integrating and licensing open-source tools I author, connecting lab protocols, publications, project knowledge, research databases, member coordination, progress tracking, and computational tools.
- Implemented local and cloud model serving for the Workbench, including specialist fine-tuned models for computational-biology workloads. Continue to expand model roles, routing, and deployment across available compute.
- Continue to develop agents that retrieve and organise knowledge, coordinate research tasks, interact with external tools and databases, and dispatch jobs to the university supercomputing facility.
- Develop modular BIOL 213 teaching workflows that assign specialist models to course modules and use session traces to assess how students reason with and use AI.
- Helped secure funding for an NVIDIA DGX Spark and local storage, establishing the compute foundation for an institution-controlled local AI capability.

INDEPENDENT ENGINEERING

September 2025 to present

Agentic-driver system

Harness-independent agentic control plane | November 2025 to present

Developing agentic-driver, a harness-independent agentic control-plane system built on integrations with Pi and Prime Agent. It is being productionised as a modular tool to complement the separate Testudo, Nuthatch, and Hillstar Orchestrator projects.

- Formal planned work is sourced from canonical Kanban and sprint boards. Human-issued, hash-bound assignment envelopes bind the task, specification, definition of done, repository, scope, capabilities, placement, and expiry before execution; the agent cannot create or self-approve its authority.
- Provides a closed-loop self-service cycle inside provisioned Testudo containment, allowing agents to report progress, access permitted state, and continue approved routine work.
- Coordinates with Testudo for Linux microVM or native macOS container execution, sanitisation, ingress and egress controls, exfiltration protection, receipt-bound output exchange, audit logging, and dangerous-command termination. Coordinates with Hillstar for workflow routing, retries, checkpoint and replay, and per-step traces.
- Routes inference between fine-tuned models served through the NVIDIA DGX Spark and approved cloud providers, while preserving structured handoff, evidence, recovery, and deterministic reconciliation.

Testudo

<https://github.com/evoclock/testudo>

Public hardened agent runtime for declarative workflows, with a FastAPI bridge, TypeScript/React renderer, input and output sanitisation, HMAC receipt-bound output exchange, append-only audit logs, permissions, and container or microVM isolation.

Nuthatch

<https://github.com/evoclock/nuthatch>

Public local-first GraphRAG system for structured scientific corpora. Backed by my proprietary clean-room implementation of a Bayesian degree-corrected stochastic block model, supporting flat and nested hierarchical inference with minimum-description-length model selection. Provides a principled alternative to modularity-optimising Leiden and Louvain methods, alongside schema-validated ingestion, embeddings, OCR, read-only MCP tools, routing keys, and token-economy measurement.

Hillstar Orchestrator

<https://github.com/evoclock/hillstar-orchestrator>

Public reproducibility-first workflow orchestrator with DAG execution, multi-provider and local-model integration, retries, checkpoint and replay, cost tracking, credential redaction, and auditable workflow traces.

Local Model Workforce and Qwen serving toolkit

<https://github.com/evoclock/local-model-workforce> · <https://github.com/evoclock/qwen-serving-toolkit>

Fine-tuned and evaluated task-bound models in the 4B to 7B class, including a 4B Implementer, and applied LoRA fine-tuning to models in the 27B to 35B class. Evaluated role-specific behaviour and coding performance through repeated controlled runs. Built model-serving utilities for bounded Qwen reasoning, vLLM integration, latency control, and safe local-service operations.

EDUCATION

Texas A&M University -- PhD Candidate, Computational Genomics

Expected completion Spring 2027

University College London -- MSci, Cell and Developmental Biology

2016